

Q14

$n = 11 =$

~~1011~~
~~1011~~
~~1011~~
~~1011~~

1101

1011
1010
1010

~~num = num & (1 << 1)~~
~~1011 & (1101)~~

$n = 11 = 1011$

sek temp naam ko variable create karo.
→ hme last bit ko check karna hai agr
0 hai toh 1 bit >> (right shift) karo
or agr 1 hai toh usko temp me
store karo.

→ last bit ko check karna ho to 1 ke saath AND
kr.

$11 = 1011 \& 1 = 0001$

~~(1011) & (0001)~~

→ dono ke par 1 hai.
To usko temp me store karo.
curr temp ++ kardo.

→ Ab next bit ko check karna hai

$11 = 1011, 1 = 0001$

dono 1 hai

temp me store karo;
temp ++ kardo;

Q14

$n = 11$

~~1011~~
~~1011~~
~~1011~~
~~1011~~

1101

1011
1010
1010

~~num = num & (1 << 1)~~

~~1011 & (1101)~~

$n = 11 = 1011$

→ ek temp naam ko variable create karo.

→ hme last bit ko check karna hai agr 0 hai toh 1 bit >> (right shift) karo or agr 1 hai toh usko temp me store karo.

→ last bit ko check karna ho to 1 ke saath AND karo.

$11 = 1011 \& 1 = 0001$

~~(1011) & (0001)~~

→ dono ke par 1 hai.

To usko temp me store karo.
aur temp ++ kardo.

→ Ab next bit ko check karna hai

$11 = 1011$, $1 = 0001$

dono 1 hai

temp me store karo;
temp ++ kardo;

Phir se next bit check karo.

$11 = 1011$, $1 = 0001$

dono different hai

→ To 11 ko right shift karo 1 se.

$11 \gg 1$
 $= 101111$

$\gg 1101$, 0001

Ab dono ~~same~~ hai.
temp me store karo
temp++. karo.

Ab Phir next bit

1101 , 0001

dono same hai
temp me store karo.

Code:-

```
int temp = 0;
while (n > 0) {
    if ((n & 1) == 1) {
        temp++;
    }
    n = n >> 1;
}
return temp;
```